

RESEARCH ARTICLE

Computational Neuroscience

Impact of sleep deprivation on aperiodic activity: a resting-state EEG study

 Duo Bai,^{1,2} Jingyi Hu,^{1,2}  Simon Jülich,^{1,2} and  Xu Lei^{1,2}

¹Sleep and Neuroimaging Center, Faculty of Psychology, Southwest University, Chongqing, China and ²Key Laboratory of Cognition and Personality (Ministry of Education), Chongqing, China

Abstract

Sleep deprivation (SD) has been shown to have a negative impact on alertness, as evidenced by behavioral and electroencephalographic studies. Nevertheless, in prior studies utilizing conventional fixed-bandwidth spectral analysis the aperiodic and periodic components were often confused, and some important periodic parameters (i.e., center frequency, bandwidth) were ignored. Here, based on a large open-access dataset of SD, we employed a standardized process for multiple-electrode analysis and group inference. We found that, compared to the healthy sleep control state (SC), the aperiodic offset shifted overall after SD, primarily in the occipital region. This shift was associated with a reduction in subjective alertness. Regarding periodic components, we did not find any power change in the alpha rhythm, but there was an increase in bandwidth of alpha within different regions distributed in the occipital and temporal lobes. These findings highlight the potential significance and value of aperiodic parameters in behavioral and electrophysiological research.

NEW & NOTEWORTHY Aperiodic and periodic components were separated in a large open-access EEG dataset of sleep deprivation. Aperiodic offsets increase after deprivation, particularly in the occipital region, reflecting a decline in self-reported vigilance. Parameterized alpha bandwidth, which was ignored in previous studies, is found to be relevant to sleep deprivation. Increase in bandwidth of alpha was focused in the occipital and temporal lobes.

alpha rhythm; aperiodic activity; electroencephalography; sleep deprivation; sleepiness

INTRODUCTION

The average human being can spend around a third of each day asleep. Sleep is one of the fundamental physiological processes necessary for human survival (1). When insufficient sleep results in diminished performance and a decline in health, sleep deprivation occurs (2). Nonetheless, laboratory sleep deprivation (SD) is an effective experimental procedure that can help clarify the connection between sleep and behavioral, emotional, and cognitive functions (3). After SD, healthy people generally feel sleepy and physically drained, and their alertness declines (4).

A great number of studies have investigated the neuroelectrophysiological response after SD. Electroencephalography (EEG) and magnetoencephalography (MEG) serve as techniques for measuring neural oscillation changes after SD. Changes in alpha band are considered to be closely related to SD (5, 6). However, the current evidence regarding how SD affects alpha rhythm is inconsistent. On one hand, Wu et al.

(7) investigated the effects of SD on resting-state alpha-band EEG, discovering reduced alpha activity, whereas other research found an increase in alpha power after deprivation (8–10). On the other hand, some studies (11) did not find that SD affects alpha power in the resting state.

Crucially, the spectrum calculated from scalp EEG signals is a combination of narrowband oscillations and broadband noise, the latter also known as aperiodic components (12). Neural oscillations are repetitive patterns of brain activity characterized by specific frequencies (i.e., alpha, 8–13 Hz). The aperiodic components, often referred to as background activity or “scale-free” broadband activity, follow a 1/f-like distribution, exhibiting decreasing spectral power as frequency increases (13). Aperiodic component activity is typically derived from two parameters: exponent (the negative slope of the power spectrum in log-log space) and offset (the uniform shift of power across frequencies) measurements. Simply put, prior research on narrowband neural oscillations has primarily examined changes within specific frequency

ranges, ignoring the contribution of “1/f” activity to the power spectra. This oversight has resulted in inaccurate interpretations of the psychological significance of these oscillations.

Considering the broadband power spectrum for SD could lead to new insights. Tramonti et al. (10) discovered a significant increase in power spectrum across all frequency bands following SD, including delta, theta, alpha, beta, and gamma. This manifests as an overall upward shift in the spectrum, which may result in increased aperiodic offset. Meanwhile, some researchers have noted that resting EEG power in the theta band was higher within the SD condition than in the control condition (SC), whereas alpha activity remained unchanged (11, 14, 15). A single night of mild sleep loss resulted in significant increases in delta, theta, and alpha power, but the difference in beta power was not statistically significant (9). After sleep deprivation, if the power of alpha remains constant while the power of theta increases and the power of beta remains unchanged but there is an increase in delta, theta, and alpha bands, the overall power distribution may exhibit a rightward skew. This could result in an increased slope (aperiodic exponent) of the spectrum.

Power spectrum analysis of EEG is widely utilized in the research of sleep and sleep-related disorders (16–18). As mentioned above, in conventional spectrum analysis the aperiodic components are often treated as noise or an interfering variable. Additionally, only changes in power are typically considered when examining alterations in neural oscillations. However, rhythmic oscillations encompass not only amplitude/power but also peak frequency and bandwidth (19). The emergence of novel strategies, fitting oscillations and the 1/f toolkit (FOOOF), has allowed researchers to parameterize neural power spectra into aperiodic components and putative periodic oscillatory activity (13). This allows the quantification of three parameterized periodic activity measures. In this research, based on FOOOF, we employed a standardized process for multiple-electrode analysis and group inference.

Recently, a study on the effect of sleep deprivation on aperiodic activity during rest has been conducted (20). However, this study included only 12 young participants and did not explore the behavioral significance of underlying changes in aperiodic components. To further investigate the psychological and behavioral implications of aperiodic and periodic components following sleep deprivation, we utilized a large open-access dataset of resting-state EEG data (21, 22). After decomposing the resting-state EEG spectrum into its periodic and aperiodic components, we explored the psychological properties of these parameters. We hypothesized, in line with previous animal studies, that SD would increase the aperiodic exponent and offset and would further influence the periodic components, i.e., alpha rhythm, which had previously been attributed to oscillatory changes.

METHODS

Participants

The resting-state EEG data of 71 participants between sleep control (SC) and SD were recorded in the Sleep and NeuroImaging Center of Southwest University, China.

Currently, this dataset can be obtained from OpenNeuro (Ref. 21; <https://openneuro.org/datasets/ds004902>). The cohort comprises 34 females and 37 males (age range: 17–23 yr, 20 ± 1.44). None of the participants had mental illness, experienced symptoms of anxiety or depression, or had sleep issues (such as insomnia). The data was shared by Xiang et al. (22) in accordance with the Declaration of Helsinki. The study was approved by the Ethics Committee of Southwest University. All procedures involved were executed in accordance with the Declaration of Helsinki.

Procedure

All participants visited the laboratory twice. The two visits, one SD and one SC session, were separated by approximately a week, and the order of participants was randomized. Before the experiment was conducted, participants underwent sleep quality monitoring using sleep diaries or actigraphs (wGT3X-BT; Actigraph, Pensacola, FL) to ensure normal and regular sleep patterns, thus ensuring that they were following a healthy and consistent sleep schedule. During the SD session, participants underwent a 24-h period of continuous wakefulness while being monitored by research assistants. Participants were not allowed to lie down, sleep, or do any vigorous physical activity. For the SC session, participants were asked to have a night of regular sleep in their dormitories.

The day following a night of SC or 24-h SD, participants finished resting-state EEG recordings and related questionnaire measurements between 0830 and 2130. Both sessions were scheduled within a fixed time frame in the morning or in the afternoon for each subject, resulting in a difference within time of day between SC and SD of <1.5 h for 58 participants (accounting for 81.59% of the total sample). Participants were instructed to keep their eyes open for 5 min while keeping blinking and eye movements to a minimum. Subsequently, they completed the Psychomotor Vigilance Test (PVT), Stanford Sleepiness Scale (SSS), the reduced version of Sleep Diary (Sleep Diary), as well as other scales. Participants were provided with detailed explanations of the study's purpose and procedures, and written informed consent was obtained before the experiment. Detailed experimental procedures and specific data are available in the article of the original experiment (22).

Questionnaires and Behavior Measurement

In our analysis, we included the results of multiple scales and behavioral tools from the original experiment. The Pittsburgh Sleep Quality Index (PSQI), a self-report assessment tool that measures sleep quality over a month, was used to assess sleep quality. The total score of the PSQI can range from 0 to 21, where higher scores indicate worse sleep quality (23). The Chinese version of PSQI has been widely used and was shown to have good psychometric properties (24).

The reduced version of Sleep Diary (Sleep Diary), a tool for self-evaluation of sleep experiences (25), was used to record the participants' self-evaluation of their sleep experiences. The Chinese version of Sleep Diary has been validated and has been successfully used in other prior studies (26). Sleep diaries are simple to use and only require a few minutes a day to complete, with questions like “When did you turn off the lights and go to bed last night?” and “Please rate your current state of alertness” (range: from 1-Sleepy to 10-Alert).

Subjective sleepiness was assessed with the Stanford Sleepiness Scale (SSS) (27). The SSS is a one-item, seven-point scale with a descriptive label for each point ranging from 1 (wide awake) to 7 (sleep onset soon), which can be used for quick estimation of sleepiness in the moment (28). Similarly to our other materials, we selected the Chinese version as the material for the present experiment (29).

Alertness was measured with the Psychomotor Vigilance Test (PVT) (30). During the PVT, participants are instructed to look at a screen centered with a square frame. As soon as a red number appears inside the square, participants have to click the mouse. The red number then reappears after a randomized time interval ranging from 2 to 10 s. In the experiment, the PVT was conducted for 5 min. The median reaction time (RT) was used to objectively assess subjects' vigilance.

Physiological Recording and Data Preprocessing

Sixty-one Ag/AgCl electrodes on an elastic cap were used to record EEG signals in accordance with the worldwide extended 10–20 electrode placement system (Brain Products GmbH, Steingrabenstr., Germany). The FCz was utilized as the online reference electrode. The sampling rate was 500 Hz.

Segments containing noise were manually rejected. Subsequently, 225.13 ± 29.17 s of SC data and 231.00 ± 38.08 s of SD data remained. In the case of an electrode with a bad fit, the signals of the defective electrode were interpolated with signals from the electrodes around it. Consequently, we interpolated 2.38 ± 2.15 electrodes in SC conditions and 2.44 ± 2.26 electrodes in SD conditions. In both steps, there were no significant differences in both the length of valid data and the number of electrodes that were found to have a bad fit (paired *t* test, $P > 0.05$).

Preprocessing of the data was conducted with MATLAB R2021b (The MathWorks) and EEGLAB (version 2021, <http://sccn.ucsd.edu/>). A standard processing pipeline was used, which included 0.2–45 Hz band filtering. Eye movement artifacts were removed from the EEG data with independent component analysis (ICA). Finally, the signals were rereferenced to average reference.

The Three-Step Standardized Process

To extend the application of SpecParam for group inference for high-density EEG, we applied a standardized analysis process that considers both multielectrode setup and group inference (as illustrated in Fig. 1). This process consists of three steps:

Step 1: Electrode-level estimation using the SpecParam model. An EEG power spectrum was calculated in each electrode by Welch's method (Hamming window: 4 s; overlap: 0). The calculated power spectral density (PSD) was then imported into Python (version 3.9.16) for further analysis. We used the SpecParam algorithm (formally known as FOOF; Ref. 13; <https://github.com/foof-tools/foof>) to parameterize power spectra into their aperiodic and periodic components. A mathematically detailed description of this algorithm can be found in the Supplemental Material (available at <https://doi.org/10.6084/m9.figshare.27085888.v1>). Consistent with previous studies (31, 32), we adopted the following settings to parameterize the spectra from 1 to 45 Hz

with a 0.25-Hz frequency resolution: peak width limits: 1–12; max number of peaks: 8; minimum peak height: 0.1; peak threshold: 2 standard deviations; and aperiodic mode: fixed. This resulted in each electrode having two aperiodic parameters: exponent and offset. Additionally, $3 \times k$ periodic parameters were used, where k (0–8) indicates the number of periodic components, each of which consists of three parameters (peak frequency, power, and bandwidth).

Step 2: Subject-level aggregation. Subjects were excluded from further analysis if more than one-third of the electrodes had a poor fit ($R^2 < 0.9$) in the SpecParam model.

Aperiodic parameters, i.e., exponent and offset, are calculated by averaging across all electrodes. Here we employed the same method as proposed by Hill et al. (33). In their study, Hill et al. introduced the concept of “global” exponent and offset in the statistical analysis, which is obtained by averaging the exponent and offset values across electrodes for each participant (33, 34). The exponent reflects the pattern of aperiodic power across frequencies and is equivalent to the negative slope of the power spectrum when measured in log-log space. There, offset reflects the uniform shift of power across frequencies. Change in both parameters may reflect in behaviorally and physiologically meaningful ways. As all electrodes contribute to these parameters, we considered exponent and offset as global parameters for the whole brain. As such, we aggregated the electrode-level data to obtain these parameters for each subject.

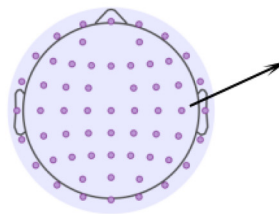
Given that there is usually more than one periodic component, to obtain the periodic parameters for each subject the component with maximum power was selected in each electrode and then averaged. In this study we focused on alpha frequency band. In the first step, we considered any component with peak frequency between 8 and 13 Hz a candidate alpha. Second, if there were several potential alpha candidates, the alpha candidate with the highest power was selected as the electrode's maximum power alpha. If in a subject's data no components with peak frequency between 8 and 13 Hz were recorded, the electrode was eliminated from further periodic components analysis because of “lack” of alpha. Second, the three parameters (peak frequency, adjusted power, and bandwidth) of the maximum power alpha were averaged across all the remaining electrodes to represent the periodic parameters of the subject. In the here-proposed framework, the periodic component of interest (here alpha) can be determined by research hypothesis. If there is no hypothesis, we suggest using the component with maximum power for a wide frequency range.

Step 3: Group-level statistics. As each subject has one group of aperiodic components (consisting of 2 parameters) and one group of large periodic components (consisting of 3 parameters), analysis is conducted for these five parameters including descriptive statistics or hypothesis test. The five parameters of interest in this study were exponent and offset of aperiodic components, peak frequency, adjusted power, and bandwidth of maximum power alpha.

Statistical Analyses

The questionnaire data and resting-state EEG were analyzed by using a paired-sample *t* test to compare differences between SC and SD. To analyze scalp distribution of the aperiodic components, we conducted Bonferroni-corrected tests

Step 1: Electrode level estimation



Electrode C6 in Sub #i			
Non-periodic	Expo0	Off0	
Periodic 1	Peak1	Amp1	Wid1
Periodic 2	Peak2	Amp2	Wid2
...			

Step 2: Subject level aggregation

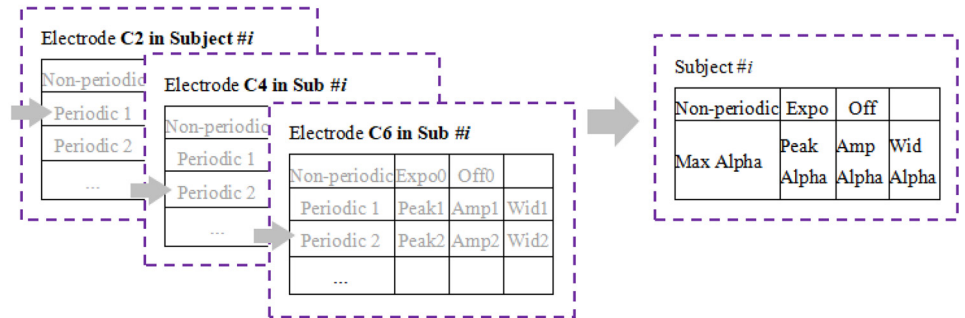
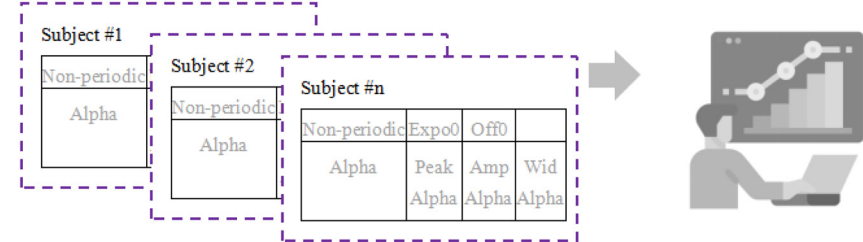


Figure 1. The flow diagram of the 3-step standardized process. *Step 1:* Electrode level estimation using SpecParam. *Step 2:* Subject-level concentration. Aperiodic parameters are averaged across all electrodes, whereas large periodic components (such as alpha rhythm) are selected from each electrode and then averaged. *Step 3:* Group-level statistics. For each subject, both aperiodic and large periodic components are obtained. Statistical analyses are then performed to derive descriptive statistics or conduct hypothesis tests.

Step 3: Group level statistics



to compare differences across electrodes between SC and SD conditions. Spearman correlation was used to further describe the relationship between SD-induced parameterized characteristics and scale scores. The parameter changes associated with SD are described as Δ , i.e., the difference between SC and SD. All statistical analyses were conducted with SPSS 25.0. Statistical significance cutoff was set at $P < 0.05$.

RESULTS

Demographic, Questionnaire, and Behavioral Characteristics

Descriptive statistics for PSQI, the Sleep Diary, median response time (RT) of PVT, and SSS in the shared dataset on SD are presented in Table 1. Because of the influence of data acquisition time, we also analyze the recording time.

As illustrated in Table 1, all 71 participants had good sleep quality during the experiment. The paired-sample t test revealed a difference in the EEG data recording time between SC and SD ($t = 2.69$, $P = 0.009$). A significant increase in SSS score after SD was observed. Meanwhile, we also noted that the RT of PVT significantly increased after SD ($t = -7.31$, $P < 0.001$). The subjective and objective manifestations of alertness (SSS and PVT) both support the effectiveness of SD.

EEG Power Spectrum

As illustrated in Fig. 2A, the energy of the whole frequency band increased after SD. Paired-sample t tests were performed between SC and SD on each frequency bin. As indexed in the bottom line, several frequency bands showed significant differences (0–9.40 Hz, 12.27–32.17 Hz, 32.84–33.39 Hz, 33.63–33.87 Hz, 34.24–35.52 Hz, 35.83–36.32 Hz) ($P < 0.05$, uncorrected) between the two conditions.

Table 1. Participants' traits and psychological behavioral data

	SC	SD	t	P
Age, yr		20.00 $\pm$ 1.44		
Sex, female%		47.89%		
PSQI		5.05 $\pm$ 2.45		
EEG record time, h:min	12:36 $\pm$ 3:08	11:55 $\pm$ 3:01	2.69	0.009
Diary sleepy	7.51 $\pm$ 1.42			
SSS	2.68 $\pm$ 1.16	4.22 $\pm$ 1.40	-4.73	<0.001
PVT (RT), ms	317.82 $\pm$ 30.85	359.03 $\pm$ 29.52	-7.31	<0.001

Data are presented as means $\pm$ standard deviation. Diary, Sleep Diary; EEG record time, the beginning time of recording EEG data; PSQI, Pittsburgh Sleep Quality Index; PVT (RT), Psychomotor Vigilance Test (median response time); SC, sleep control; SD, sleep deprivation. SSS, Stanford Sleepiness Scale. The amount of data used in analysis was as follows: PSQI, 92.96% ($n = 66$); Sleep Diary, 74.65% ($n = 53$); SSS, 52.11% ($n = 37$); and PVT, 42.25% ($n = 30$), respectively. Significant results are presented in bold ($P < 0.05$).

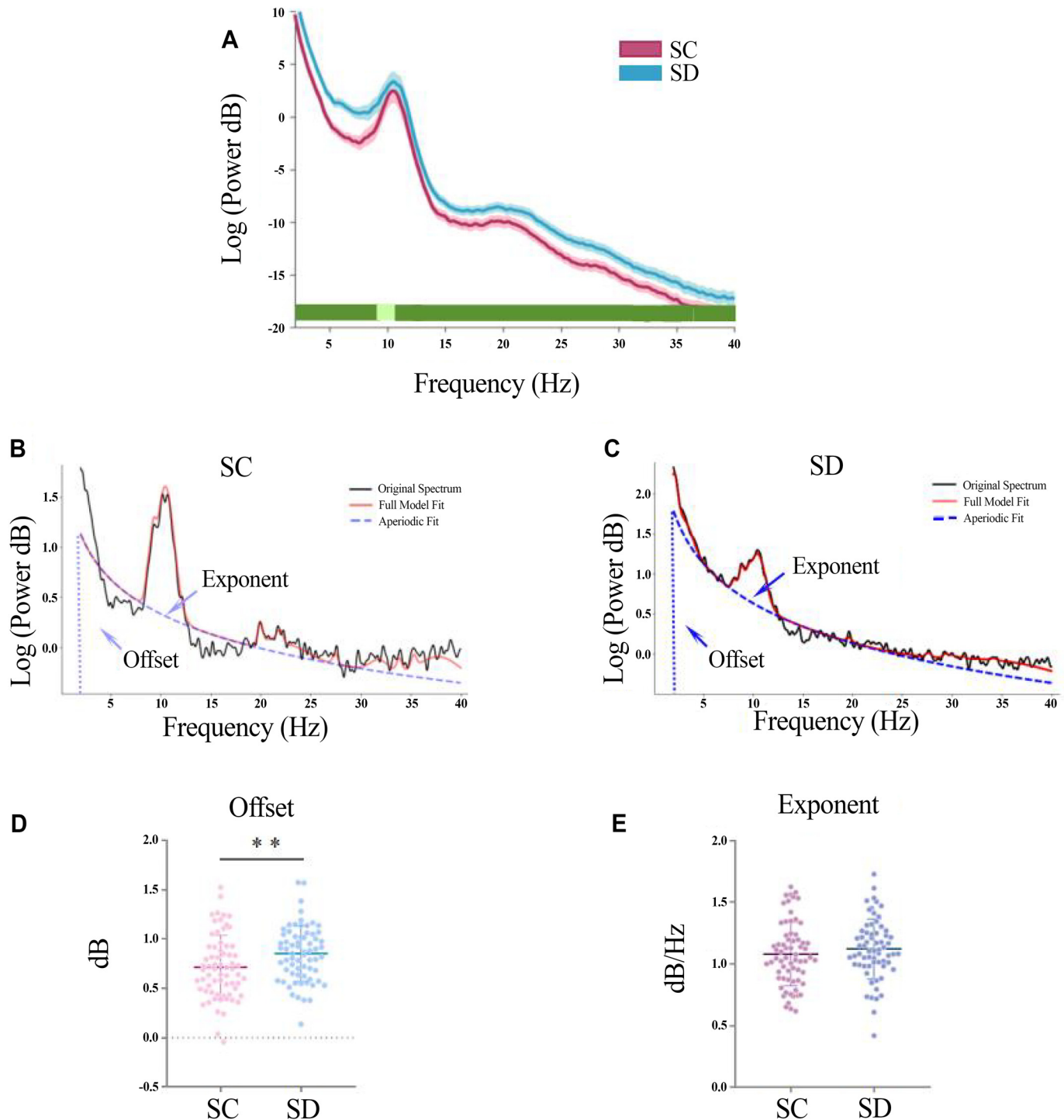


Figure 2. The total power within the range of the conventional power spectral analysis and the fit of aperiodic parameters under conditions of sleep control (SC) and sleep deprivation (SD). **A:** EEG spectrum, estimated via the conventional power spectral analysis. EEG activity measures in **A** are taken without parameterization. Center lines indicate the mean power for SC and SD, and the lighter colored areas indicate standard deviation. The green bottom band showcases the significance levels of the paired t test, with sleep conditions as factor. Different shades of green represent levels of significance (where dark green corresponds to $P < 0.05$ and light green corresponds to $P \geq 0.05$). **B:** single-subject (Sub#48) example for FOOOF model fit, with Fp2 electrode, showing the aperiodic exponent and offset (marked in blue) in SC. **C:** single-subject (Sub#48) example for FOOOF model fit, with Fp2 electrode, showing the aperiodic exponent and offset (marked in blue) in SD. **D** and **E:** offset (**D**) and exponent (**E**) parameters at subject level. $**P < 0.001$.

Aperiodic Activity

For aperiodic fitting, four subjects were excluded because of poor fit (subject numbers: Sub#10, Sub#25, Sub#28, Sub#40).

In the remaining 67 subjects, spectral parameterization of the scalp EEG data for each participant was generally well fitted by the SpecParam algorithm. Mean fit across the whole scalp for each participant was 0.98 ± 0.01 for SC and 0.98 ± 0.01 for

SD. Figure 2, B and C, display the aperiodic fits at the single-subject single-electrode level for various sleep conditions (Fp2 in Sub#48, SC: $R^2 = 0.95$, SD: $R^2 = 0.98$).

When the conditions are compared, changes in the aperiodic components appear mainly in the offset. SD led to a significant increase in offset ($t = -3.73$, $P < 0.001$), but no difference in aperiodic exponent between SC and SD ($t = -1.34$, $P = 0.18$) could be found, as shown in Fig. 2, D and E. To quantify the magnitude of differences between SC and SD, we additionally calculated Cohen's d . More detailed values are shown in Table 2.

The aperiodic parameters at the electrode level were averaged for all participants in two conditions. As illustrated in Fig. 3, topographies show the spatial patterns of aperiodic parameters between SC and SD. In both conditions, the offsets and exponents display a roughly symmetrical distribution. Specifically, offset values are larger in the frontal and occipital regions, whereas exponent values show a relatively widespread distribution, with maximal signal strength close to the vertex.

In addition, we analyzed the spatial patterns of the differences between sleep conditions (SD – SC) and highlighted the electrodes with significant differences on the topology by red circles (P1, PO3, Oz, O1, P8, POz, PO4, PO8, O2). Figure 3, C and D, demonstrate clearly that the Δ offset is primarily observed in the posterior occipital regions following the multiple comparison correction based on Bonferroni correction.

Parameterized Alpha Peak Characteristics

By subtracting the aperiodic components from the raw PSD, the SpecParam algorithm revealed the periodic characteristics of the power spectrum. As illustrated in Table 2 and Fig. 4, the bandwidth of the periodic alpha components considerably increased after sleep loss ($t = -2.50$, $P = 0.015$). There were no significant differences in the center frequency of the alpha peak between the conditions ($t = -0.02$, $P = 0.99$). Even though there was a decreasing trend in the adjusted power of alpha, it did not reach statistical significance ($t = 1.44$, $P = 0.15$). The ground-average periodic alpha parameter distribution on the scalp is depicted in Fig. 4, D–F. Figure 4F demonstrates that the variation in bandwidth is primarily manifested in the occipital and temporal lobes.

Correlations Between Behavior and EEG Parameters

Another key point of this study was to further explore the relationship between 1/f offset, periodic alpha activity, and cognitive behavioral performance. These findings are presented in Fig. 5. After deprivation, Δ offset was negatively correlated with

the dimension “difference in sleep diary alertness” ($r = -0.33$, $P = 0.019$, $n = 51$; Fig. 5A), and marginally associated with the differences in SSS scores ($r = 0.35$, $P = 0.05$, $n = 32$; Fig. 5B). We did not find any correlation between Δ offset and PVT_RT ($r = 0.21$, $P = 0.28$). Additionally, we found no correlation between the increase in periodic alpha bandwidth and questionnaire or behavior following deprivation. As mentioned above, there was a significant difference in collection time between SC and SD. Additional correlation analysis was conducted, and recording time was not significantly correlated with scale behavior (Diary, $r = 0.04$, $P = 0.79$; SSS, $r = -0.04$, $P = 0.82$; PVT_RT, $r = -0.19$, $P = 0.34$).

DISCUSSION

This work analyzed the aperiodic and periodic components of scalp EEG, using a large public dataset of sleep deprivation. By employing the SpecParam algorithm, we found that aperiodic offsets increased after SD, particularly in the posterior occipital regions. When aperiodic activity was removed, the bandwidth of the parameterized alpha components showed an increase following SD, whereas the center frequency and adjusted power remained constant. Changes in aperiodic offset were linked to a decline in self-reported alertness but were not correlated with objective measure of alertness. Overall, our findings demonstrate that alterations in aperiodic and periodic parameters following SD, as well as their behavioral significance, confirm the reformation of parameterized algorithms for EEG spectral calculation.

Sleep Deprivation Altered Aperiodic Offset

Our results show that offsets increased significantly after total SD and that these shifts in aperiodic parameters mainly are reflected in the occipital areas. No changes were observed in the aperiodic exponents due to sleep loss. Consistent with previous studies, we observed that the spatial distribution of aperiodic parameters was roughly symmetrical (see Fig. 3), with offsets greater in frontal and occipital areas and exponents larger in the parietal lobe (31, 35). The consistent specific scalp distribution of the aperiodic components highlights the functional significance of scale-free brain activity rather than unstructured noise (36).

Lendner et al. (20) were the first to report the effect of SD on aperiodic activity during rest. They recorded resting-state EEG data from 12 young individuals after SD and SC. They observed a broadband increase in power across frequencies after SD, which is consistent with our findings. However, they attributed the increase in spectral power after SD to an increase in the aperiodic exponents. From the PSD in their

Table 2. The results of SpecParam model fits within different sleep conditions

	SC	SD	t	P	Cohen's d
<i>Aperiodic components</i>					
Exponent, dB/Hz	1.08 ± 0.26	1.12 ± 0.24	−1.34	0.18	0.16
Offset, dB	0.72 ± 0.32	0.85 ± 0.28	−3.73	<0.001	0.46
<i>Periodic components of the alpha peak (8–13 Hz)</i>					
Center frequency, Hz	10.72 ± 0.78	10.72 ± 0.90	−0.02	0.99	0.00
Adjusted power, dB	0.74 ± 0.38	0.69 ± 0.35	1.44	0.15	0.18
Bandwidth, Hz	1.60 ± 0.40	1.74 ± 0.46	−2.50	0.015	0.31

Data are presented as means ± standard deviation. SC, sleep control; SD, sleep deprivation. Cohen's d was taken as absolute value. Significant results are presented in bold ($P < 0.05$).

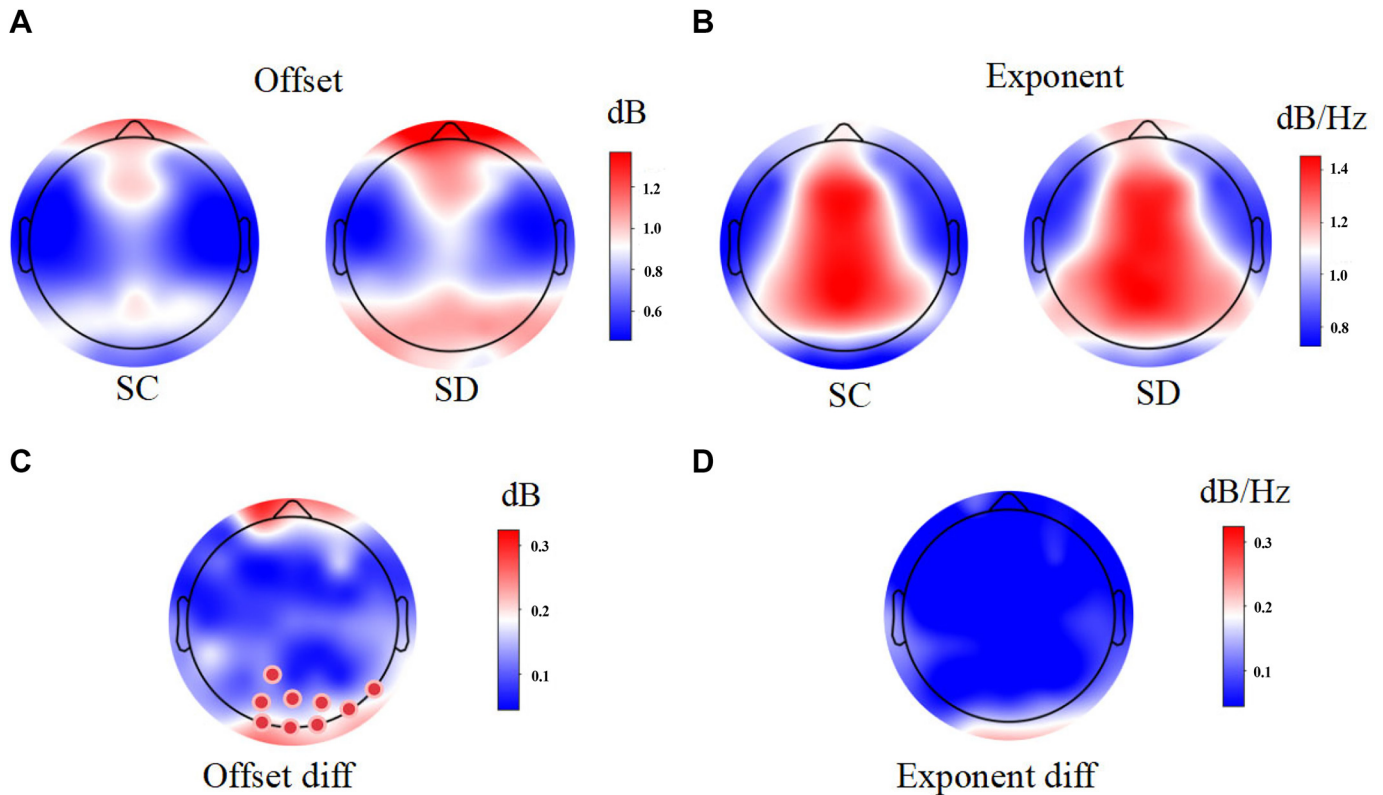


Figure 3. The distribution of scalp topography in sleep control (SC) and sleep deprivation (SD) conditions. *A*: topographical maps showcasing aperiodic offset for all electrodes in SC and SD. *B*: the aperiodic exponent for all electrodes in SC and SD. *C*: comparisons of aperiodic offset between SC and SD. Paired-sample *t* tests on each electrode of the scalp between SC and SD conditions were conducted. Electrodes with $P < 0.05$ are marked with red dots (Bonferroni corrected). *D*: comparisons of aperiodic exponent between SC and SD.

work, appears it that there is elevated energy in the 0–8 Hz frequency band after SD; this is very different from our PSD. Additionally, the small sample size of the study of Lendner et al. could have influenced the manifestation of aperiodic results (37).

Manning et al. (38) noted that the aperiodic offset can directly reflect the rate of spiking activity. Sleep could facilitate a regulatory decrease in firing rate (39). Prolonged wakefulness and SD have been proven to enhance abnormal spiking activity in neurons (firing rates and the number of active neurons), even if sleep restriction was only brief (40, 41), as confirmed by numerous animal studies (42–44). Changes in neural spike activity induced by SD contribute to local field potentials (LFPs), which constitute large parts of the EEG signal (45, 46). In vivo measurements observed that SD increased neural spike activity in the cortex of rats (47). At the same time, there was a discernible rise in the delta and theta band power of LFP. According to Campbell et al. (48), there is a trade-off between increased cortical reactivity and decreased long-term potentiation (LTP). However, according to our findings, an increase in aperiodic offset may indicate abnormal neuronal spiking in cortical neurons after SD, which is accompanied by a decrease in signal transmission between pre- and postsynaptic neurons. We discovered that SD-driven spatial variability of the aperiodic components is particularly located within the occipital regions. Consistent with prior research, we observed a robust association between the offset and the degree of

blood oxygen dependence both in the resting state and during task execution (49–51). We also observed that the offset reflected increased activity in the bilateral cerebellum and cingulate gyrus during eyes-open resting-state tasks, while being associated with decreased blood oxygen level-dependent (BOLD) activity in structures such as the cingulate gyrus, bilateral occipital gyri, and middle frontal gyrus (51). More concretely, SD is linked to an increase in aperiodic offset, which partly reflects the decrease in occipital lobe activity.

In a systematic review of 95 functional (f)MRI studies of SD by Zheng et al. (52), the occipital gyrus was identified as a brain region that may be directly involved in the neurophysiological abnormalities induced by SD. One night of total SD has been shown to reduce activity in the occipital visual cortex (53–55) and functional connectivity between the occipital cortex and other regions (8, 56–59). This study again validates the value of aperiodic offset in elucidating abnormal brain mechanics associated with SD.

Exponents may represent the balance of excitatory and inhibitory neuronal activity (E:I balance) (60, 61) and are linked to research on expensive and invasive techniques for achieving E:I equilibrium via gamma-aminobutyric acid (GABA) and glutamate expression (62). A steeper slope is indicative of increased inhibitory neural activity and/or decreased excitatory activity, whereas a flatter slope is associated with increased excitatory activity and/or decreased inhibitory activity (63). Contrary to the hypothesis, we did

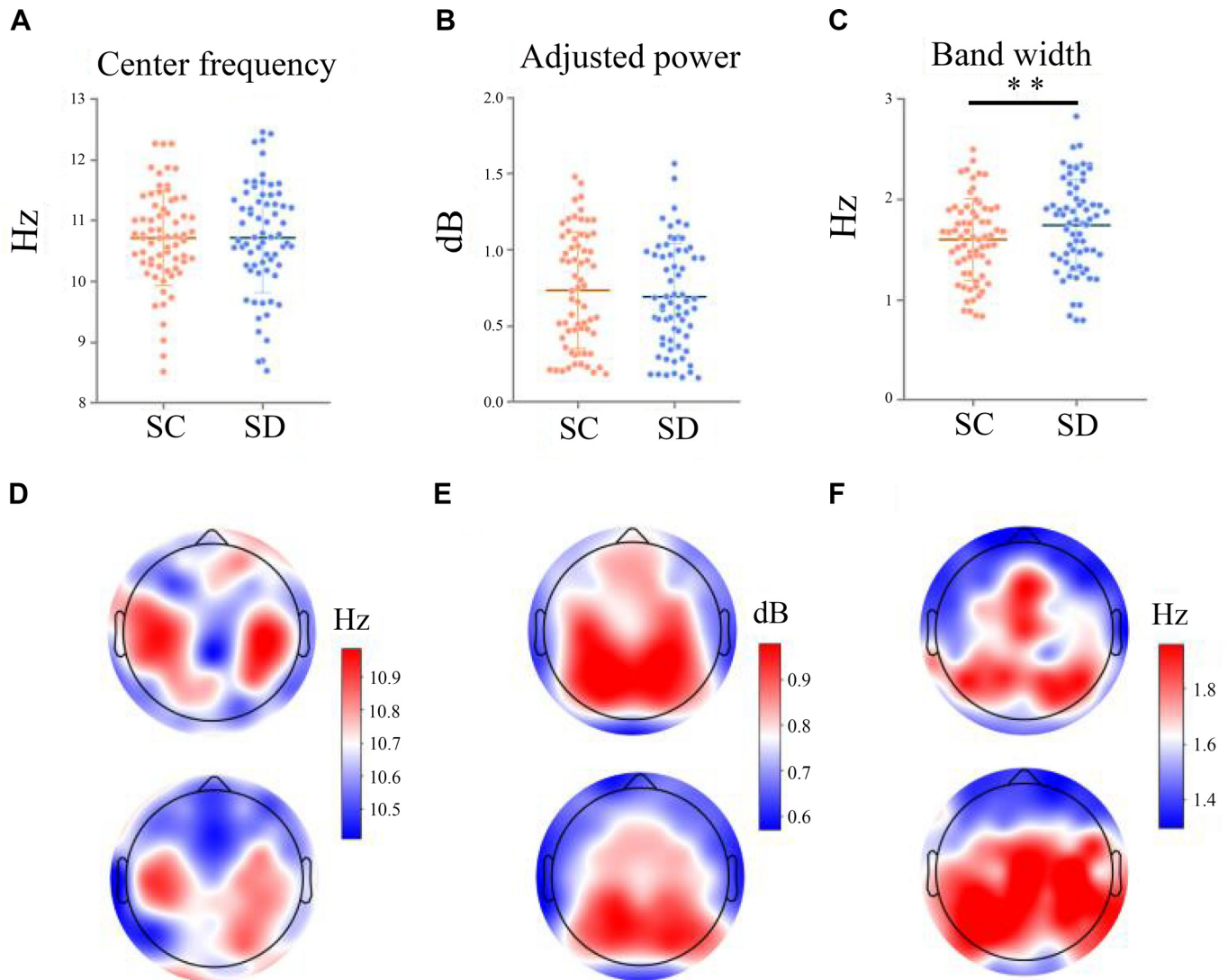
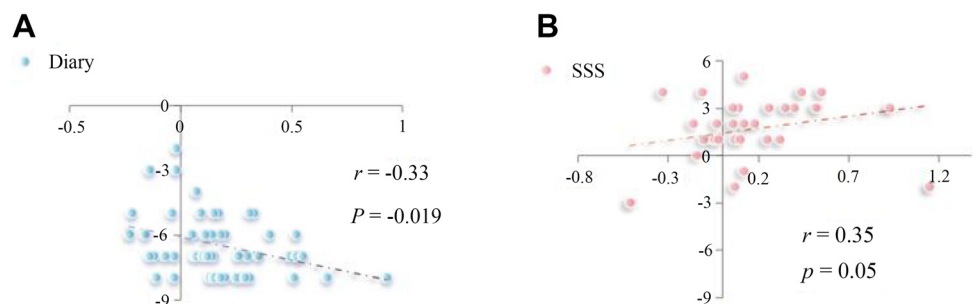


Figure 4. Differences in parameterized alpha activity and scalp distribution under various sleep conditions. *A*: differences in the center frequency characteristics of periodic alpha between sleep control (SC) and sleep deprivation (SD). *B*: variations in adjusted power of periodic alpha between SC and SD. *C*: differences in periodic alpha bandwidth features between SC and SD. *D–F*: the corresponding scalp distributions (top: SC; bottom: SD).

not note any changes in aperiodic exponents. It is possible that after deprivation there is an increase in neuronal instability (64), which better reflects the increase in inhibitory neurons, especially in the local cortex. Yet, when considering the interaction of excitatory and inhibitory neurons throughout the brain, this change is obscured. Prolonged wakefulness in rodents seems to lead to “local sleep”

symptoms in an otherwise awake brain, which are accompanied by a slow/theta wave (2–6 Hz) in the local EEG that occurs at various times in various cortical areas (65). Additionally, research examining the test-retest properties of aperiodic activity indicates that offset was substantially more stable and consistent than the exponent and that the FOOOF method exhibited a higher goodness of fit (32, 66). The higher

Figure 5. Relationship between EEG aperiodic parameters and subjective cognitive. *A*: scatterplot of $\Delta 1/f$ offset with the difference value of sleep diary alertness between sleep control (SC) and sleep deprivation (SD). The larger the offset, the lower the alertness dimension scores of the sleep diary ($r = -0.33$, $P = 0.019$, $n = 51$). *B*: the correlation between Δ offset and Stanford Sleepiness Scale (SSS) scores. Increased offset may be accompanied by an increase in reported drowsiness ($r = 0.35$, $P = 0.05$, $n = 32$).



stability of the offset allows us to explore changes between sleep control and sleep deprivation over a 1-wk interval, effectively controlling for the temporal variable.

Sleep Deprivation Altered Alpha Activity

We initially applied conventional spectral analysis to examine changes in the power within specific frequency bands (8–13 Hz) between SC and SD. Alpha power increased significantly after sleep deprivation (6, 10). Although these results broadly reflect expected patterns of activity, prior approaches did not correct for this effect to discern whether the difference arises from changes in the periodic or the aperiodic components (13). Underlying aperiodic activity may result in misinterpretation of periodic activity. This likely contributes to the conflicting findings of periodic alpha activity assessment by sleep loss (9, 11). Additionally, conventional analysis often ignores variations in peak frequencies and bandwidths within oscillation bands (67, 68).

We explored the differences in parameterized alpha components (center frequency, adjusted power, bandwidth) between SC and SD. When accounting for aperiodic component activity, SD is related to increase in alpha bandwidth. Wider bandwidth suggests the involvement of decentralized dominant frequency in alpha rhythm activity. Results from topographical maps indicate that after deprivation the increase in alpha bandwidth is primarily observed in the occipital and temporal cortical regions. This may imply an increased instability of neural activity around these channels. SD affects the perceptual transitions, and the stability of an individual perceptual representation is at least partially mediated by the simultaneously induced increase in alpha power following deprivation (69). Moreover, the estimated generators of the alpha rhythm controlling perceptual stability were circumscribed and located in the high-level visual cortex, suggesting that the influence of alpha rhythm operates at a high level in the visual hierarchy. The positive impact of alpha power on stabilizing perception may be attributed to its modulation of alpha bandwidth. However, we did not observe any changes in center frequency. Although the adjusted alpha power showed a decreasing trend after deprivation, it did not reach statistical significance. Additionally, we found no link between the change in alpha bandwidth and sleepiness or vigilance. The variations in alpha bandwidth may have additional implications that we did not detect, as the range of psychological behavioral measurements in this study was limited. Overall, the results of the parameterization algorithm indicate that previous differences in the alpha power may be artifactual and driven by differences in the aperiodic exponent and offset, which obscures the real alpha activity (70, 71), including peak frequency and bandwidth.

Aperiodic Activity and Cognitive Behavior in Sleep Deprivation

Additionally, we investigated the psychophysiological significance of aperiodic and periodic parameters. In the present study, an increase in aperiodic offset following sleep deprivation corresponded to a decrease in self-rated vigilance and could be slightly associated with higher levels of sleepiness. SD increased the level of cortical responsiveness

in the brain (10). Changes in voltage distribution on the scalp were found to be positively correlated with subjective and behavioral measures of sleepiness (72). Stojanoski et al. (73) argued that even a slight sleep loss on a single night greatly decreases attentiveness and increased sleepiness. In this study, this was again demonstrated by decreased evoked potential activity of the occipital visual cortex during PVT. The increase in offset after deprivation reflected a rise in cortical responsiveness, which could be an abnormal signal indicating occipital lobe activity impairment, resulting in decreased vigilance and increased drowsiness. Nevertheless, we did not find a correlation between the aperiodic offset and the RT of PVT. This suggests that offset differences primarily reflect a “feeling” of decreased alertness and are insufficient to reflect on objective alertness. Torrado et al. (39) demonstrated a segregation in the homeostatic regulation of neuronal firing between sleep and wakefulness. Downward homeostatic regulation is restricted to sleep, whereas upward homeostatic regulation only occurs during wakefulness. In other words, the increase in aperiodic offsets following deprivation may result from the upregulation of neurons during wakefulness, which is only reflected in subjective perception. Objective alertness, however, may be more closely associated with downward homeostatic regulation. Recent studies have shown that aperiodic activity exhibits less significant objective task-related changes than oscillatory activity (74).

Limitations

There are several limitations in this study that should be considered. First, the resting-state EEG data from sleep-deprived individuals that we used for our analysis contained very limited cognitive testing (22). As only subject and object vigilance were tested in this dataset, and only reaction time in the PVT was considered, our exploration of the aperiodic psychophysiological implications is preliminary. Second, although we conducted additional calculations to determine the effect sizes of the differences between the aperiodic and periodic components of SC and SD, according to Cohen's *d*, it is evident that both offset and bandwidth exhibit only a moderate effect size. Third, while calculating individualized periodic alpha activity, we noted that some electrodes did not contain alpha components, which may have impacted the representativeness of the final alpha parameters obtained. However, in this study the percentage of electrodes that contained alpha components was still above 82%, which should be sufficiently high. Fourth, we only utilized the alpha band to provide an example, which we used to compare the results of conventional spectral calculation and parametric algorithm. Other rhythms should be considered for analysis in future research. In this study, we decided to focus on the alpha rhythm, as it has been shown to be closely related to SD (5), and as previous findings on the impact of SD on alpha rhythm activity have been inconsistent, we decided that this research should focus on alpha rhythm activity, to help in clearing up those inconsistencies.

Conclusions

This study utilized an open-access dataset of sleep deprivation. We observed an increase in aperiodic offsets, particularly in the occipital region. The increases in offsets

were associated with a decline in self-reported vigilance following deprivation, which was partially indicative of an increase in sleepiness. However, the change in offset may not reflect any objective changes in alertness, potentially because of the segregation of neuronal homeostatic regulation between wakefulness and sleep periods. When considering the periodic components, there was an increase in the bandwidth of parameterized alpha, indicating instability of neural activity in occipital and temporal lobes. Collectively, these findings highlight the potential significance and value of aperiodic parameters in behavioral and electrophysiological research.

DATA AVAILABILITY

The data that support the findings of this study are openly available in OpenNeuro at <https://openneuro.org/datasets/ds004902>.

SUPPLEMENTAL MATERIAL

Supplemental Material: <https://doi.org/10.6084/m9.figshare.27085888.v1>.

GRANTS

This research was supported by grants from National Natural Science Foundation of China (32471095) and National Key Research and Development Program of China (2021YFC2501500).

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

D.B., J.H., and X.L. conceived and designed research; X.L. performed experiments; D.B. and J.H. analyzed data; D.B. interpreted results of experiments; D.B. prepared figures; D.B. drafted manuscript; S.J. and X.L. edited and revised manuscript; D.B. and X.L. approved final version of manuscript.

REFERENCES

- Irwin MR. Why sleep is important for health: a psychoneuroimmunology perspective. *Annu Rev Psychol* 66: 143–172, 2015. doi:10.1146/annurev-psych-010213-115205.
- Abrams RM. Sleep deprivation. *Obstet Gynecol Clin North Am* 42: 493–506, 2015. doi:10.1016/j.ogc.2015.05.013.
- Pilcher JJ, Huffcutt AI. Effects of sleep deprivation on performance: a meta-analysis. *Sleep* 19: 318–326, 1996. doi:10.1093/sleep/19.4.318.
- Killgore WD. Effects of sleep deprivation on cognition. *Prog Brain Res* 185: 105–129, 2010. doi:10.1016/B978-0-444-53702-7.00007-5.
- Strijkstra AM, Beersma DG, Drayer B, Halbesma N, Daan S. Subjective sleepiness correlates negatively with global alpha (8–12 Hz) and positively with central frontal theta (4–8 Hz) frequencies in the human resting awake electroencephalogram. *Neurosci Lett* 340: 17–20, 2003. doi:10.1016/S0304-3940(03)00033-8.
- Gibbings A, Ray LB, Berberian N, Nguyen T, Shahidi Zandi A, Owen AM, Comeau FJ, Fogel SM. EEG and behavioural correlates of mild sleep deprivation and vigilance. *Clin Neurophysiol* 132: 45–55, 2021. doi:10.1016/j.clinph.2020.10.010.
- Wu J, Zhou Q, Li J, Chen Y, Shao S, Xiao Y. Decreased resting-state alpha-band activation and functional connectivity after sleep deprivation. *Sci Rep* 11: 484, 2021. doi:10.1038/s41598-020-79816-8.
- Zhang L, Huang Y, Zhang Y, Xin W, Shao Y, Yang Y. Enhanced high-frequency precuneus-cortical effective connectivity is associated with decreased sensory gating following total sleep deprivation. *NeuroImage* 197: 255–263, 2019. doi:10.1016/j.neuroimage.2019.04.057.
- Gibbings A, Ray LB, Gagnon S, Collin CA, Robillard R, Fogel SM. The EEG correlates and dangerous behavioral consequences of drowsy driving after a single night of mild sleep deprivation. *Physiol Behav* 252: 113822, 2022. doi:10.1016/j.physbeh.2022.113822.
- Tramonti Fantozzi MP, Banfi T, Di Galante M, Ciuti G, Faraguna U. Sleep deprivation-induced changes in baseline brain activity and vigilant attention performance. *Brain Sci* 12: 1690, 2022. doi:10.3390/brainsci12121690.
- Magnuson JR, Kang HJ, Dalton BH, McNeil CJ. Neural effects of sleep deprivation on inhibitory control and emotion processing. *Behav Brain Res* 426: 113845, 2022. doi:10.1016/j.bbr.2022.113845.
- Barry RJ, De Blasio FM. Characterizing pink and white noise in the human electroencephalogram. *J Neural Eng* 18, 2021. doi:10.1088/1741-2552/abe399.
- Donoghue T, Haller M, Peterson EJ, Varma P, Sebastian P, Gao R, Noto T, Lara AH, Wallis JD, Knight RT, Shestuyk A, Voytek B. Parameterizing neural power spectra into periodic and aperiodic components. *Nat Neurosci* 23: 1655–1665, 2020. doi:10.1038/s41593-020-00744-x.
- Hoedlmoser K, Griessenberger H, Fellingner R, Freunberger R, Klimesch W, Gruber W, Schabus M. Event-related activity and phase locking during a psychomotor vigilance task over the course of sleep deprivation. *J Sleep Res* 20: 377–385, 2011. doi:10.1111/j.1365-2869.2010.00892.x.
- Bernardi G, Siclari F, Yu X, Zennig C, Bellesi M, Ricciardi E, Cirelli C, Ghilardi MF, Pietrini P, Tononi G. Neural and behavioral correlates of extended training during sleep deprivation in humans: evidence for local, task-specific effects. *J Neurosci* 35: 4487–4500, 2015. doi:10.1523/JNEUROSCI.4567-14.2015.
- Zhao W, Van Someren EJ, Li C, Chen X, Gui W, Tian Y, Liu Y, Lei X. EEG spectral analysis in insomnia disorder: a systematic review and meta-analysis. *Sleep Med Rev* 59: 101457, 2021. doi:10.1016/j.smrv.2021.101457.
- Chikhi S, Matton N, Blanchet S. EEG power spectral measures of cognitive workload: a meta-analysis. *Psychophysiology* 59: e14009, 2022. doi:10.1111/psyp.14009.
- Jaramillo-Jimenez A, Tovar-Rios DA, Ospina JA, Mantilla-Ramos YJ, Loaiza-López D, Henao Isaza V, Zapata Saldarriaga LM, Cadavid Castro V, Suarez-Revelo JX, Bocanegra Y, Lopera F, Pineda-Salazar DA, Tobón Quintero CA, Ochoa-Gomez JF, Borda MG, Aarsland D, Bonanni L, Brønneck K. Spectral features of resting-state EEG in Parkinson's disease: a multicenter study using functional data analysis. *Clin Neurophysiol* 151: 28–40, 2023. doi:10.1016/j.clinph.2023.03.363.
- Bazanov OM, Vernon D. Interpreting EEG alpha activity. *Neurosci Biobehav Rev* 44: 94–110, 2014. doi:10.1016/j.neubiorev.2013.05.007.
- Lendner JD, Niethard N, Mander BA, van Schalkwijk FJ, Schuh-Hofer S, Schmidt H, Knight RT, Born J, Walker MP, Lin JJ, Helfrich RF. Human REM sleep recalibrates neural activity in support of memory formation. *Sci Adv* 9: ead1895, 2023. doi:10.1126/sciadv.ad1895.
- Chuoqin X, Xinrui F, Duo B, Ke L, Xu L. A resting-state EEG dataset for sleep deprivation. *OpenNeuro*, 2023. doi:10.18112/openneuro.ds004902.v1.0.0.
- Xiang C, Fan Bai XD, Lv K, Lei X. A resting-state EEG dataset for sleep deprivation. *Sci Data* 11: 427, 2024. doi:10.1038/s41597-024-03268-2.
- Buyssse DJ, Reynolds CF 3rd, Monk TH, Berman SR, Kupfer DJ. The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research. *Psychiatry Res* 28: 193–213, 1989. doi:10.1016/0165-1781(89)90047-4.
- Liu XC, Tang MQ. Reliability and validity of the Pittsburgh Sleep Quality Index. *Chin J Psychiatry* 29: 29103–29107, 1996.
- van Hees VT, Sabia S, Anderson KN, Denton SJ, Oliver J, Catt M, Abell JG, Kivimäki M, Trenell MI, Singh-Manoux A. A novel, open access method to assess sleep duration using a wrist-worn accelerometer. *PLoS One* 10: e0142533, 2015. doi:10.1371/journal.pone.0142533.
- Wang H, Tian Y, Wang Y, He Q, Qiu J, Feng T, Chen H, Lei X. Distinct neural responses of morningness and eveningness chronotype to homeostatic sleep pressure revealed by resting-state functional magnetic resonance imaging. *CNS Neurosci Ther* 28: 1439–1446, 2022. doi:10.1111/cns.13887.

27. Hoddes E, Zarcone V, Smythe H, Phillips R, Dement WC. Quantification of sleepiness: a new approach. *Psychophysiology* 10: 431–436, 1973. doi:10.1111/j.1469-8986.1973.tb00801.x.
28. Pantesco EJ, Kan IP. Racial and ethnic disparities in self-reported sleep duration: roles of subjective socioeconomic status and sleep norms. *Sleep Med* 112: 246–255, 2023. doi:10.1016/j.sleep.2023.10.025.
29. Wu Y, Lei Y, Chen P, Hu G, Lin B, Zhang C, Wu X, Wang L. Dissociable brainstem functional connectivity changes correlate with objective and subjective vigilance decline after total sleep deprivation in healthy male subjects. *J Neurosci Res* 101: 1044–1057, 2023. doi:10.1002/jnr.25182.
30. Balkin TJ. Behavioral biomarkers of sleepiness. *J Clin Sleep Med* 7: S12–S15, 2011. doi:10.5664/JCSM.1344.
31. Wang Z, Mo Y, Sun Y, Hu K, Peng C, Zhang S, Xue S. Separating the aperiodic and periodic components of neural activity in Parkinson's disease. *Eur J Neurosci* 56: 4889–4900, 2022. doi:10.1111/ejn.15774.
32. McKeown DJ, Finley AJ, Kelley NJ, Cavanagh JF, Keage HA, Baumann O, Schinazi VR, Moustafa AA, Angus DJ. Test-retest reliability of spectral parameterization by 1/f characterization using SpecParam. *Cereb Cortex* 34: bhad482, 2024. doi:10.1093/cercor/bhad482.
33. Hill AT, Clark GM, Bigelow FJ, Lum JA, Enticott PG. Periodic and aperiodic neural activity displays age-dependent changes across early-to-middle childhood. *Dev Cogn Neurosci* 54: 101076, 2022. doi:10.1016/j.dcn.2022.101076.
34. Zhang C, Stock AK, Mückschel M, Hommel B, Beste C. Aperiodic neural activity reflects metacontrol. *Cereb Cortex* 33: 7941–7951, 2023. doi:10.1093/cercor/bhad089.
35. Merkin A, Sghirripa S, Graetz L, Smith AE, Hordacre B, Harris R, Pitcher J, Semmler J, Rogasch NC, Goldsworthy M. Do age-related differences in aperiodic neural activity explain differences in resting EEG alpha? *Neurobiol Aging* 121: 78–87, 2023. doi:10.1016/j.neurobiolaging.2022.09.003.
36. He BJ, Zempel JM, Snyder AZ, Raichle ME. The temporal structures and functional significance of scale-free brain activity. *Neuron* 66: 353–369, 2010. doi:10.1016/j.neuron.2010.04.020.
37. Pani SM, Saba L, Fraschini M. Clinical applications of EEG power spectra aperiodic component analysis: a mini-review. *Clin Neurophysiol* 143: 1–13, 2022. doi:10.1016/j.clinph.2022.08.010.
38. Manning JR, Jacobs J, Fried I, Kahana MJ. Broadband shifts in local field potential power spectra are correlated with single-neuron spiking in humans. *J Neurosci* 29: 13613–13620, 2009. doi:10.1523/JNEUROSCI.2041-09.2009.
39. Torrado Pacheco A, Botorff J, Gao Y, Turrigiano GG. Sleep promotes downward firing rate homeostasis. *Neuron* 109: 530–544.e6, 2021. doi:10.1016/j.neuron.2020.11.001.
40. Vyazovskiy VV, Olcese U, Lazimy YM, Faraguna U, Esser SK, Williams JC, Cirelli C, Tononi G. Cortical firing and sleep homeostasis. *Neuron* 63: 865–878, 2009. doi:10.1016/j.neuron.2009.08.024.
41. Marmelshtein A, Eckerling A, Hadad B, Ben-Eliyahu S, Nir Y. Sleep-like changes in neural processing emerge during sleep deprivation in early auditory cortex. *Curr Biol* 33: 2925–2940.e6, 2023. doi:10.1016/j.cub.2023.06.022.
42. Coenen AM, Van Luijckelaar EL. Genetic animal models for absence epilepsy: a review of the WAG/Rij strain of rats. *Behav Genet* 33: 635–655, 2003. doi:10.1023/A:1026179013847.
43. Borbély S, Világi I, Haraszti Z, Szalontai Ö, Hajnik T, Tóth A, Détári L. Sleep deprivation decreases neuronal excitability and responsiveness in rats both in vivo and ex vivo. *Brain Res Bull* 137: 166–177, 2018. doi:10.1016/j.brainresbull.2017.12.003.
44. Hashizume M, Ito R, Hojo Y, Yanagawa Y, Murakoshi T. Acute sleep deprivation reduces oscillatory network inhibition in the young rat basolateral amygdala. *Neuroscience* 401: 73–83, 2019. doi:10.1016/j.neuroscience.2019.01.001.
45. Katzner S, Nauhaus I, Benucci A, Bonin V, Ringach DL, Carandini M. Local origin of field potentials in visual cortex. *Neuron* 61: 35–41, 2009. doi:10.1016/j.neuron.2008.11.016.
46. Musall S, von Pföstl V, Rauch A, Logothetis NK, Whittingstall K. Effects of neural synchrony on surface EEG. *Cereb Cortex* 24: 1045–1053, 2014. doi:10.1093/cercor/bhs389.
47. Lu Z, Xu S, Wang H, He E, Liu J, Dai Y, Xie J, Song Y, Wang Y, Wang Y, Qu L, Cai X. PNPt/MWCNT-PEDOT:PSS-modified micro-electrode arrays for the synchronous dopamine and neural spike detection in rat models of sleep deprivation. *ACS Appl Bio Mater* 4: 4872–4884, 2021. doi:10.1021/acsabm.1c00172.
48. Campbell IG, Guinan MJ, Horowitz JM. Sleep deprivation impairs long-term potentiation in rat hippocampal slices. *J Neurophysiol* 88: 1073–1076, 2002. doi:10.1152/jn.2002.88.2.1073.
49. Winawer J, Kay KN, Foster BL, Rauschecker AM, Parvizi J, Wandell BA. Asynchronous broadband signals are the principal source of the BOLD response in human visual cortex. *Curr Biol* 23: 1145–1153, 2013. doi:10.1016/j.cub.2013.05.001.
50. Lei X, Wang Y, Yuan H, Chen A. Brain scale-free properties in awake rest and NREM sleep: a simultaneous EEG/fMRI study. *Brain Topogr* 28: 292–304, 2015. doi:10.1007/s10548-014-0399-x.
51. Jacob MS, Roach BJ, Sargent KS, Mathalon DH, Ford JM. Aperiodic measures of neural excitability are associated with anticorrelated hemodynamic networks at rest: a combined EEG-fMRI study. *NeuroImage* 245: 118705, 2021. doi:10.1016/j.neuroimage.2021.118705.
52. Zheng S, Feng S, Yao H, Dong L, Feng Z, Liu X, Zhang B, Jia H, Ning Y. Altered functional connectivity after acute sleep deprivation reveals potential locations for noninvasive brain stimulation techniques. *Sleep Med* 110: 212–219, 2023. doi:10.1016/j.sleep.2023.08.019.
53. Chee MW, Goh CS, Namburi P, Parimal S, Seidl KN, Kastner S. Effects of sleep deprivation on cortical activation during directed attention in the absence and presence of visual stimuli. *NeuroImage* 58: 595–604, 2011. doi:10.1016/j.neuroimage.2011.06.058.
54. Poudel GR, Innes CR, Jones RD. Distinct neural correlates of time-on-task and transient errors during a visuomotor tracking task after sleep restriction. *NeuroImage* 77: 105–113, 2013. doi:10.1016/j.neuroimage.2013.03.054.
55. Feng S, Yao H, Zheng S, Feng Z, Liu X, Liu R, Dong L, Cai Y, Jia H, Ning Y. Altered functional connectivity in working memory network after acute sleep deprivation. *Neuroscience* 535: 158–167, 2023. doi:10.1016/j.neuroscience.2023.11.003.
56. Alonso JF, Romero S, Mañanas MA, Alcalá M, Antonijoan RM, Giménez S. Acute sleep deprivation induces a local brain transfer information increase in the frontal cortex in a widespread decrease context. *Sensors (Basel)*, 16: 540, 2016. doi:10.3390/s16040540.
57. Fu W, Dai C, Chen J, Wang L, Song T, Peng Z, Xu M, Xu L, Tang Y, Shao Y. Altered insular functional connectivity correlates to impaired vigilant attention after sleep deprivation: a resting-state functional magnetic resonance imaging study. *Front Neurosci* 16: 889009, 2022. doi:10.3389/fnins.2022.889009.
58. Chuah LY, Chee MW. Functional neuroimaging of sleep deprived healthy volunteers and persons with sleep disorders: a brief review. *Ann Acad Med Singap* 37: 689–694, 2008. doi:10.47102/annals-acadmedsg.V37N8p689.
59. Javaheripour N, Shahdipour N, Noori K, Zarei M, Camilleri JA, Laird AR, Fox PT, Eickhoff SB, Eickhoff CR, Rosenzweig I, Khazale H, Tahmasian M. Functional brain alterations in acute sleep deprivation: an activation likelihood estimation meta-analysis. *Sleep Med Rev* 46: 64–73, 2019. doi:10.1016/j.smrv.2019.03.008.
60. Bódizs R, Szalárdy O, Horváth C, Ujma PP, Gombos F, Simor P, Pótári A, Zeising M, Steiger A, Dresler M. A set of composite, non-redundant EEG measures of NREM sleep based on the power law scaling of the Fourier spectrum. *Sci Rep* 11: 2041, 2021. doi:10.1038/s41598-021-81230-7.
61. Waschke L, Donoghue T, Fiedler L, Smith S, Garrett DD, Voytek B, Obleser J. Modality-specific tracking of attention and sensory statistics in the human electrophysiological spectral exponent. *eLife* 10: e70068, 2021. doi:10.7554/eLife.70068.
62. Gao R, Peterson EJ, Voytek B. Inferring synaptic excitation/inhibition balance from field potentials. *NeuroImage* 158: 70–78, 2017. doi:10.1016/j.neuroimage.2017.06.078.
63. Finley AJ, Angus DJ, van Reekum CM, Davidson RJ, Schaefer SM. Periodic and aperiodic contributions to theta-beta ratios across adulthood. *Psychophysiology* 59: e14113, 2022 [Erratum in *Psychophysiology* 61: e14555, 2024]. doi:10.1111/psyp.14113.
64. Vyazovskiy VV, Olcese U, Hanlon EC, Nir Y, Cirelli C, Tononi G. Local sleep in awake rats. *Nature* 472: 443–447, 2011. doi:10.1038/nature10009.
65. Hung CS, Sarasso S, Ferrarelli F, Riedner B, Ghilardi MF, Cirelli C, Tononi G. Local experience-dependent changes in the wake EEG after prolonged wakefulness. *Sleep* 36: 59–72, 2013. doi:10.5665/sleep.2302.

66. **Li N, Yang J, Long C, Lei X.** Test-retest reliability of EEG aperiodic components in resting and mental task states. *Brain Topogr* 37: 961–971, 2024. doi:[10.1007/s10548-024-01067-x](https://doi.org/10.1007/s10548-024-01067-x).
67. **Karekal A, Stuart S, Mancini M, Swann NC.** Elevated Gaussian-modeled beta power in the cortex characterizes aging, but not Parkinson's disease. *J Neurophysiol* 129: 1086–1093, 2023. doi:[10.1152/jn.00480.2022](https://doi.org/10.1152/jn.00480.2022).
68. **Leroy S, Major S, Bublitz V, Dreier JP, Koch S.** Unveiling age-independent spectral markers of propofol-induced loss of consciousness by decomposing the electroencephalographic spectrum into its periodic and aperiodic components. *Front Aging Neurosci* 14: 1076393, 2022. doi:[10.3389/fnagi.2022.1076393](https://doi.org/10.3389/fnagi.2022.1076393).
69. **Piantoni G, Romeijn N, Gomez-Herrero G, Van Der Werf YD, Van Someren EJ.** Alpha power predicts persistence of bistable perception. *Sci Rep* 7: 5208, 2017. doi:[10.1038/s41598-017-05610-8](https://doi.org/10.1038/s41598-017-05610-8).
70. **Chiang AK, Rennie CJ, Robinson PA, van Albada SJ, Kerr CC.** Age trends and sex differences of alpha rhythms including split alpha peaks. *Clin Neurophysiol* 122: 1505–1517, 2011. doi:[10.1016/j.clinph.2011.01.040](https://doi.org/10.1016/j.clinph.2011.01.040).
71. **Haegens S, Cousijn H, Wallis G, Harrison PJ, Nobre AC.** Inter- and intra-individual variability in alpha peak frequency. *NeuroImage* 92: 46–55, 2014. doi:[10.1016/j.neuroimage.2014.01.049](https://doi.org/10.1016/j.neuroimage.2014.01.049).
72. **Gorgoni M, Ferlazzo F, Ferrara M, Moroni F, D'Atri A, Fanelli S, Gizzi Torriglia I, Lauri G, Marzano C, Rossini PM, De Gennaro L.** Topographic electroencephalogram changes associated with psychomotor vigilance task performance after sleep deprivation. *Sleep Med* 15: 1132–1139, 2014. doi:[10.1016/j.sleep.2014.04.022](https://doi.org/10.1016/j.sleep.2014.04.022).
73. **Stojanoski B, Benoit A, Van Den Berg N, Ray LB, Owen AM, Shahidi Zandi A, Quddus A, Comeau FJ, Fogel SM.** Sustained vigilance is negatively affected by mild and acute sleep loss reflected by reduced capacity for decision making, motor preparation, and execution. *Sleep* 42: zsy200, 2019. doi:[10.1093/sleep/zsy200](https://doi.org/10.1093/sleep/zsy200).
74. **Wainio-Theberge S, Wolff A, Gomez-Pilar J, Zhang J, Northoff G.** Variability and task-responsiveness of electrophysiological dynamics: scale-free stability and oscillatory flexibility. *NeuroImage* 256: 119245, 2022. doi:[10.1016/j.neuroimage.2022.119245](https://doi.org/10.1016/j.neuroimage.2022.119245).